Visión por Computador

Versión 2026



Stable Difusion

(CAPÍTULO 3 – CV031)



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How to generate images from text?

Key Idea of CLIP

Two people relax at a beach under a tree, enjoying a serene ocean view.

Text Encoder

$\mathbf{x}_1$

EMBEDDING 1

$$sc(\mathbf{x}_1, \mathbf{x}_2) = \mathbf{x}_1 \mathbf{x}_2^T \rightarrow \text{HIGH}$$

(must be) VERY SIMILAR



Image Encoder

$\mathbf{x}_2$

EMBEDDING 2

Key Idea of Text to Image Generation

Two people relax at a beach under a tree, enjoying a serene ocean view.

Text Encoder

$\mathbf{x}_1$

EMBEDDING 1

$$sc(\mathbf{x}_1, \mathbf{x}_2) = \mathbf{x}_1 \mathbf{x}_2^T \rightarrow \text{HIGH}$$

(must be) VERY SIMILAR



Image Encoder

$\mathbf{x}_2$

EMBEDDING 2

Deep Unsupervised Learning using Nonequilibrium Thermodynamics

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Abstract

A central problem in machine learning involves modeling complex data-sets using highly flexible families of probability distributions in which learning, sampling, inference, and evaluation are still analytically or computationally tractable. Here, we develop an approach that simultaneously achieves both flexibility and tractability. The essential idea, inspired by non-equilibrium these models are unable to aptly describe structure in rich datasets. On the other hand, models that are *flexible* can be molded to fit structure in arbitrary data. For example, we can define models in terms of any (non-negative) function $\phi(\mathbf{x})$ yielding the flexible distribution $p(\mathbf{x}) = \frac{\phi(\mathbf{x})}{Z}$, where Z is a normalization constant. However, computing this normalization constant is generally intractable. Evaluating, training, or drawing samples from such flexible models typically requires a very expensive Monte Carlo process.

Image Generation – How to generate an image from noise?

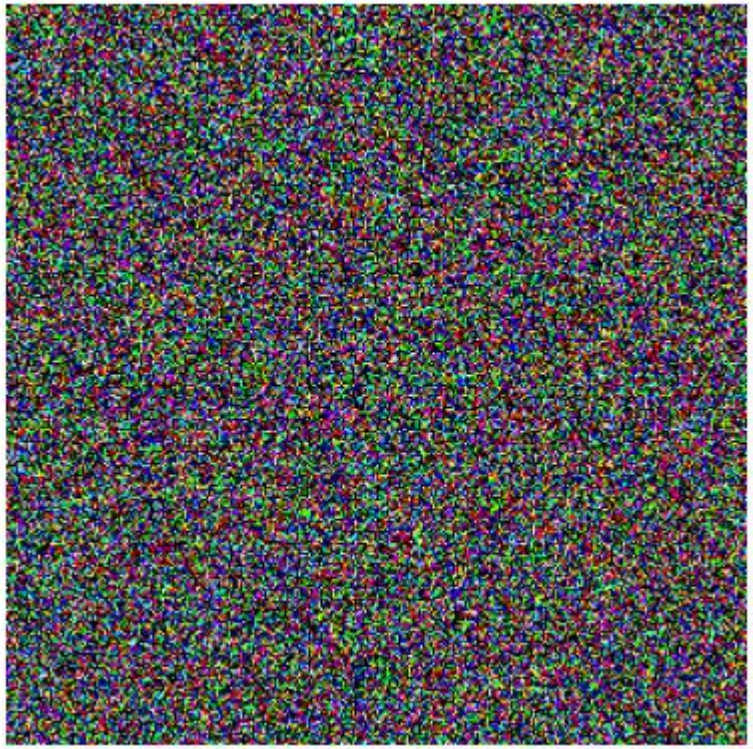
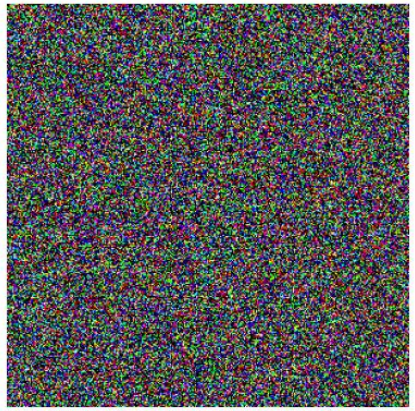




Image Generation – From Image to Noise





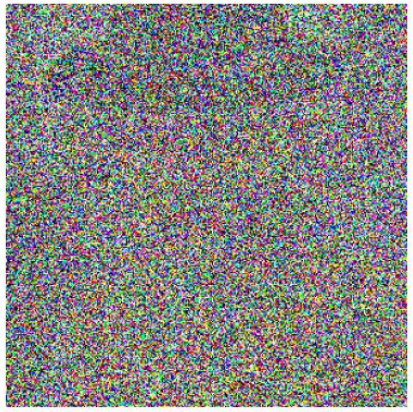


Image Generation – Idea

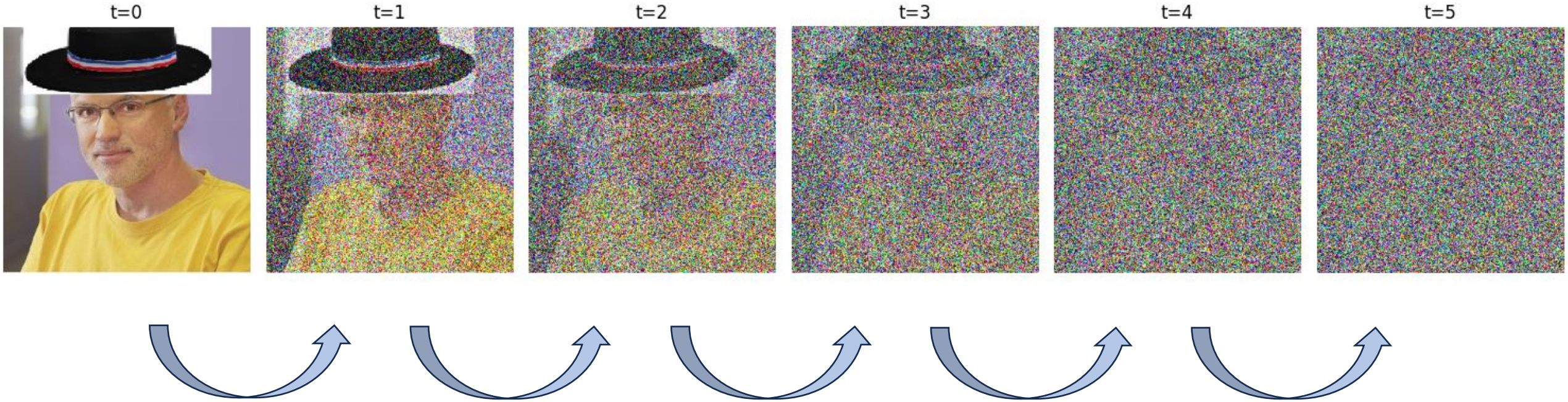
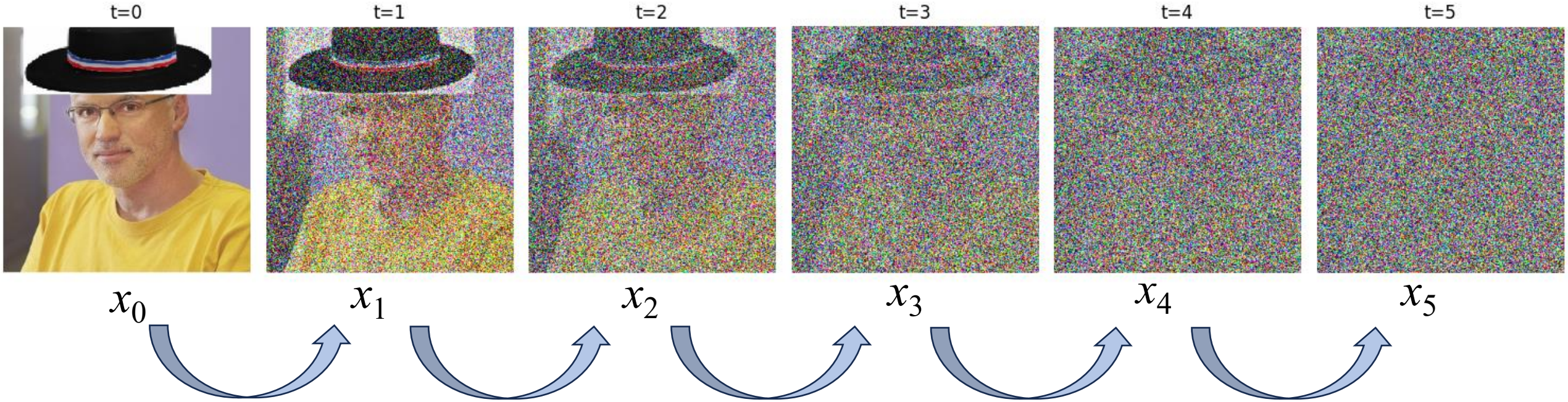


Image Generation – Idea

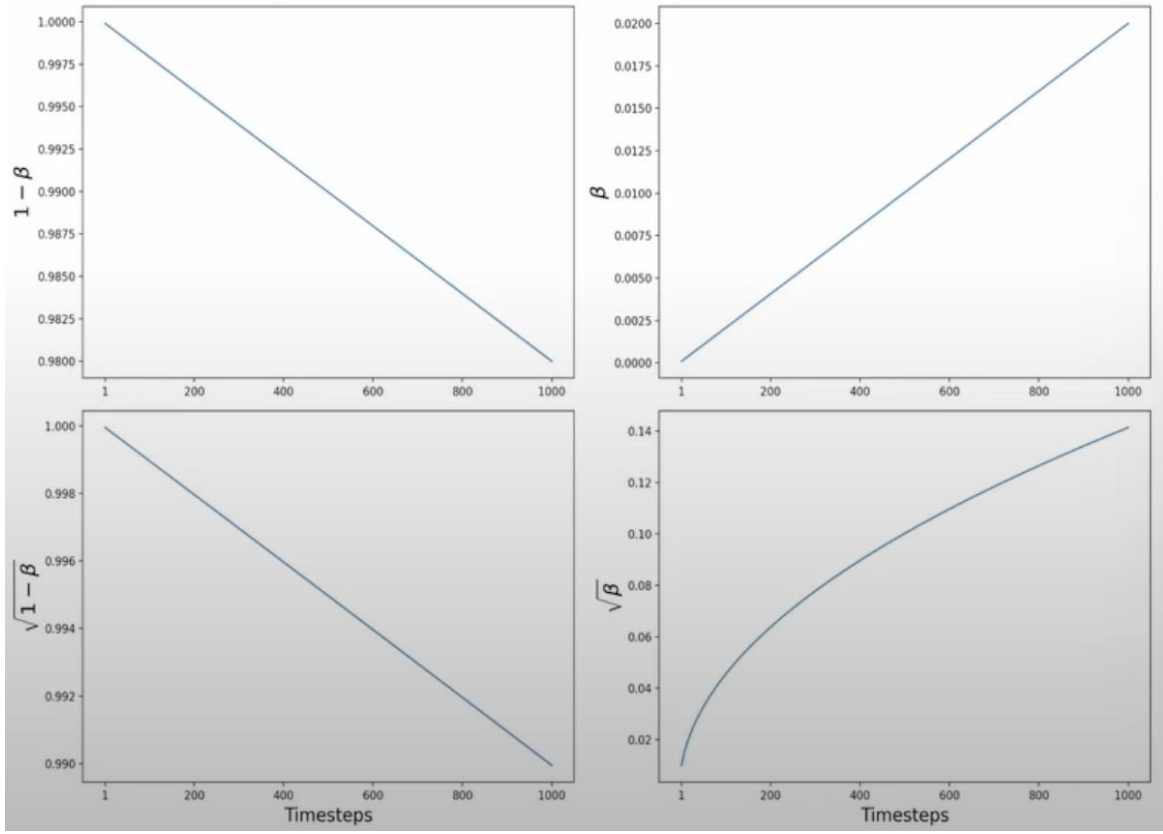


noise

$$x_t = \sqrt{1-\beta_t}\, x_{t-1} + \sqrt{\beta_t}\,\epsilon$$

$$; \text{where } \epsilon \sim \mathcal{N}(0, I)$$

β_t Amount of Gaussian noise added at timestep t



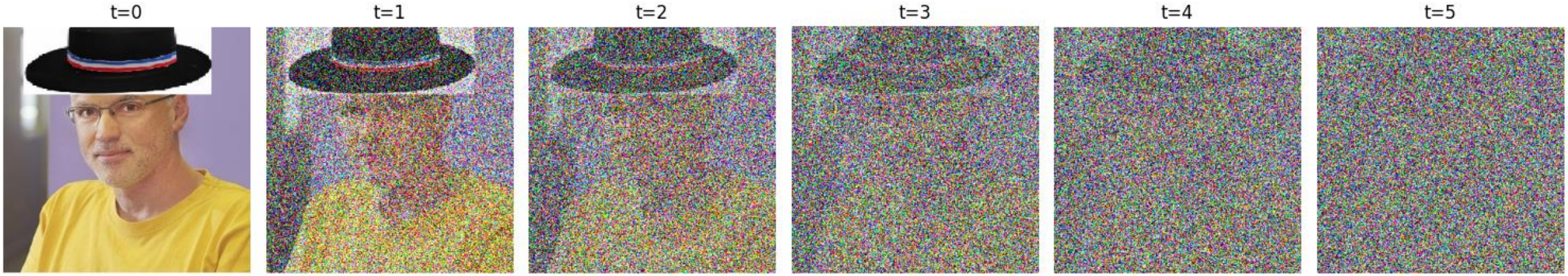
By increasing Beta we control the amount of noise that gets added to the image at various timesteps

$$x_t = \sqrt{1-\beta_t}\, x_{t-1} + \sqrt{\beta_t}\, \epsilon$$

$$; \text{where } \epsilon \sim \mathcal{N}(0, I)$$

β_t Amount of Gaussian noise added at timestep t

Image Generation – Idea

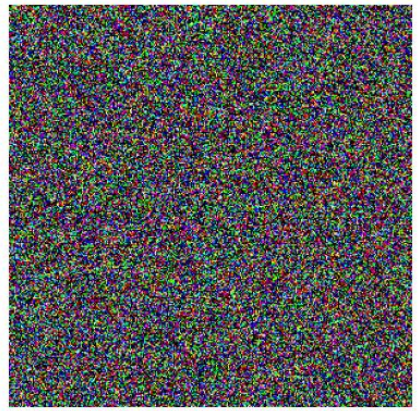


x_0 x_1 x_2 x_3 x_4 x_5

noise

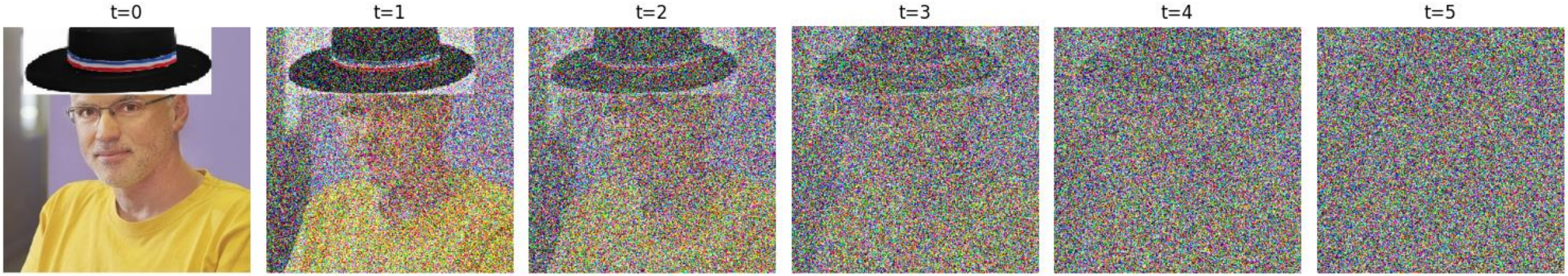
$$x_t = \sqrt{1-\beta_t}\, x_{t-1} + \sqrt{\beta_t}\,\epsilon$$

$$; \text{where } \epsilon \sim \mathcal{N}(0, I)$$



β_t Amount of Gaussian noise added at timestep t

Image Generation – Idea



$$\alpha_t := 1 - \beta_t$$

$$\bar{\alpha}_t := \prod_{s=1}^{t} \alpha_s$$

$$x_t := \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1 - \bar{\alpha}_t}\, \epsilon$$

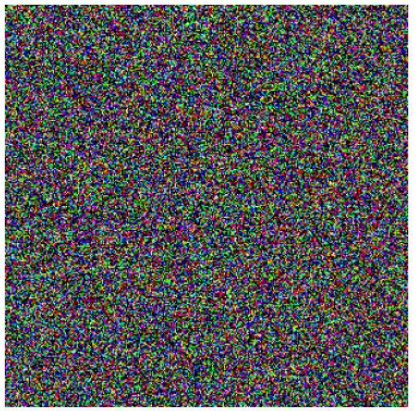


Image Generation – Idea

$$t = 2$$



$$x_t := \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t}\,\epsilon$$



Image Generation – Idea

$$t = 1$$



$$x_t := \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t}\,\epsilon$$

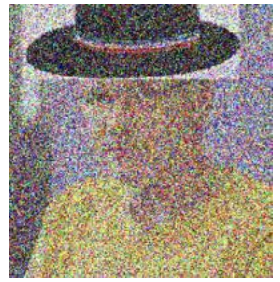


Image Generation – Idea

$$t = 4$$



$$x_t := \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t}\,\epsilon$$

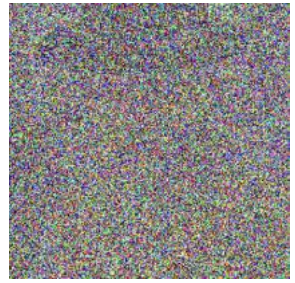


Image Generation – Idea : UNet for noise estimation

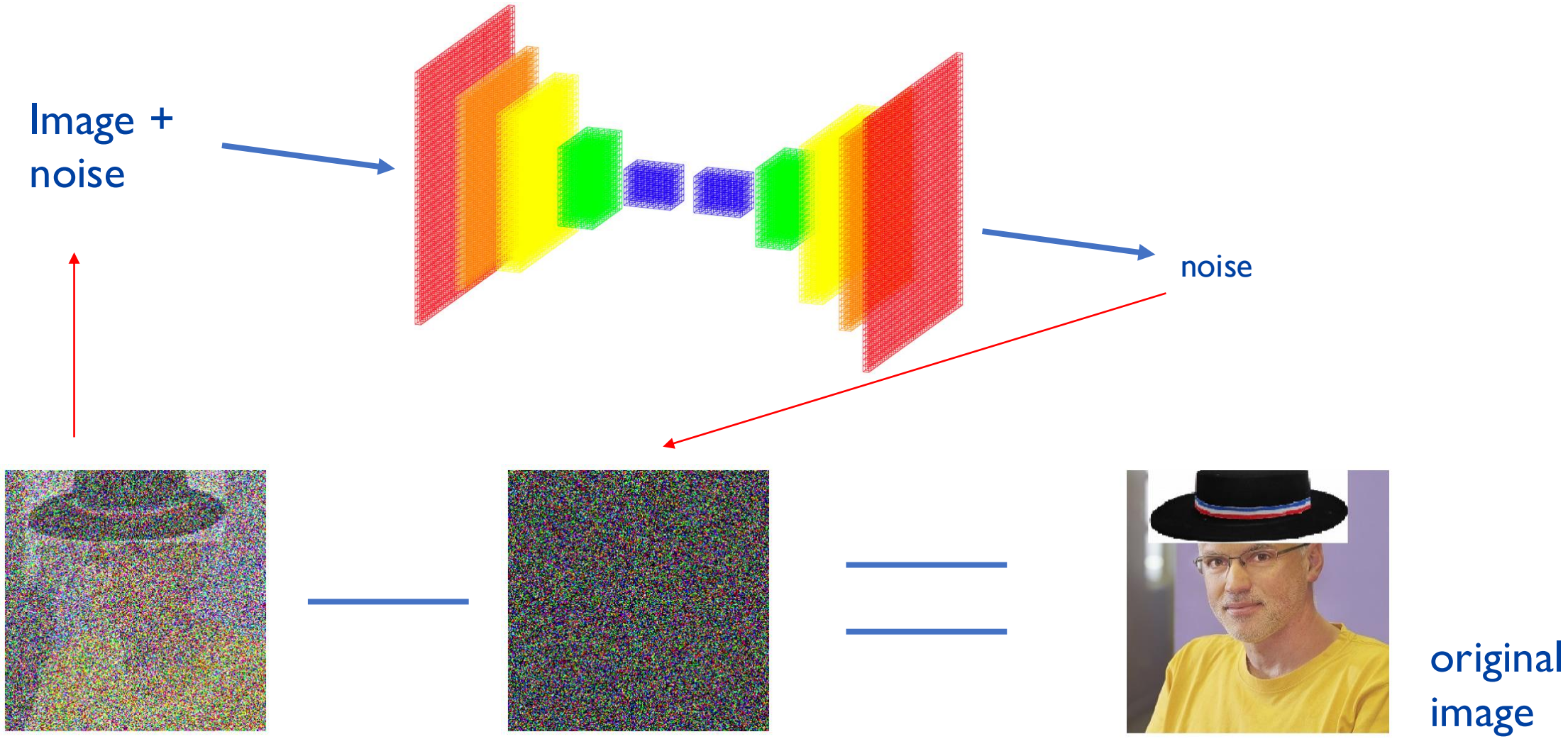
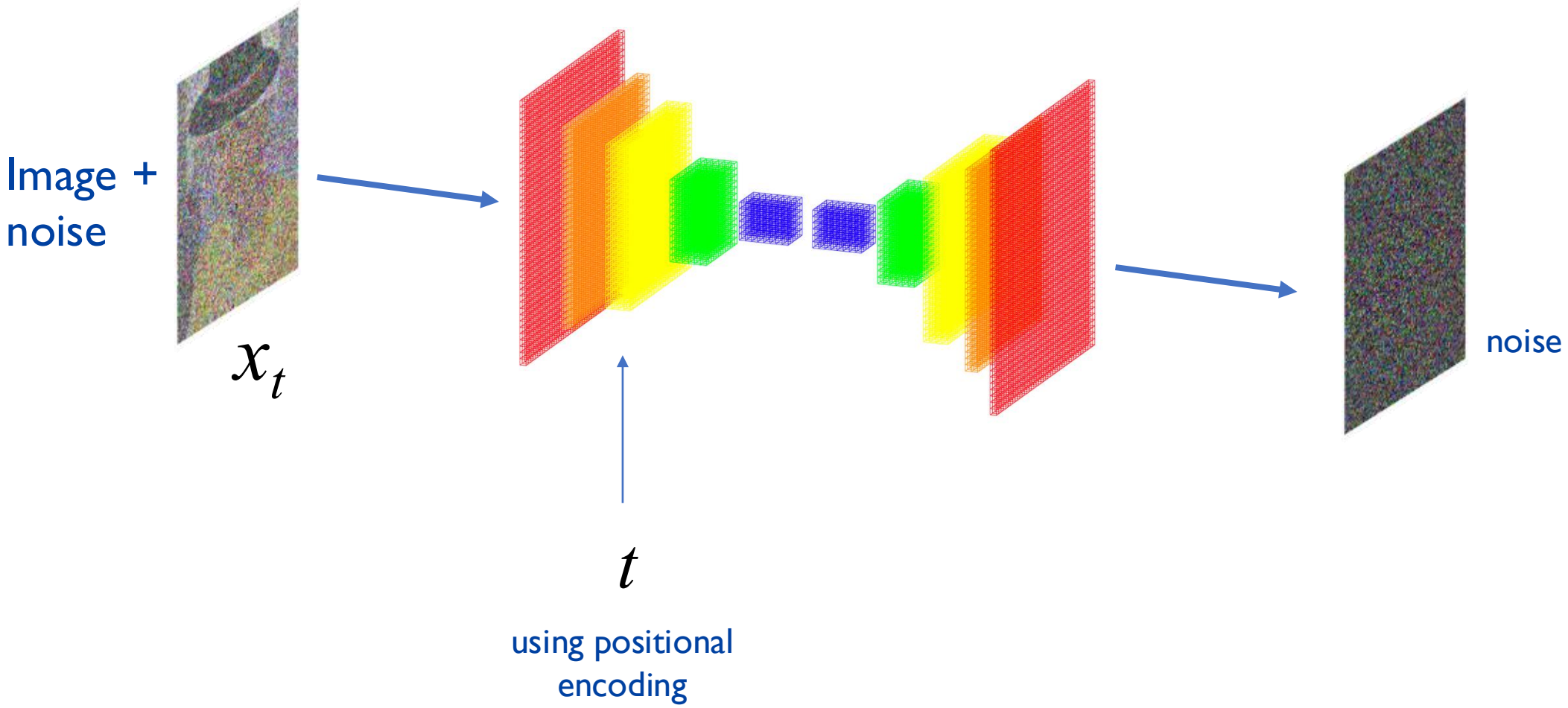


Image Generation – Idea : UNet for noise estimation



Training

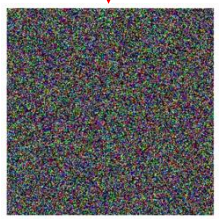


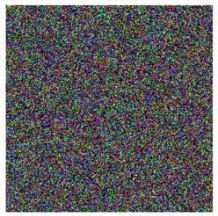
Algorithm 1 Training

1: **repeat**
2: $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
3: $t \sim \text{Uniform}(\{1, \dots, T\})$
4: $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
5: Take gradient descent step on

$$\nabla_\theta \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t) \right\|^2$$

6: **until** converged







Select image

Choose random t

Gaussian noise

Predict noise $\boldsymbol{\epsilon}_\theta$ and optimize using gradient descent relative to actual noise t

Testing: Reverse diffusion

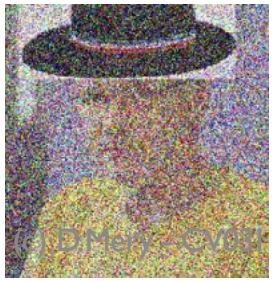
Algorithm 2 Sampling

1: $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
2: **for** $t = T, \ldots, 1$ **do**
3: $\quad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\mathbf{z} = \mathbf{0}$
4: $\quad \mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$
5: **end for**
6: **return** $\mathbf{x}_0$

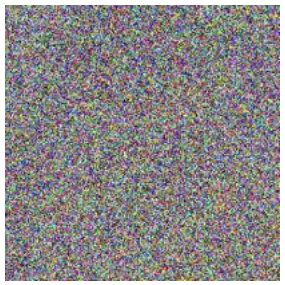
Random noise

Radom noise for t>1

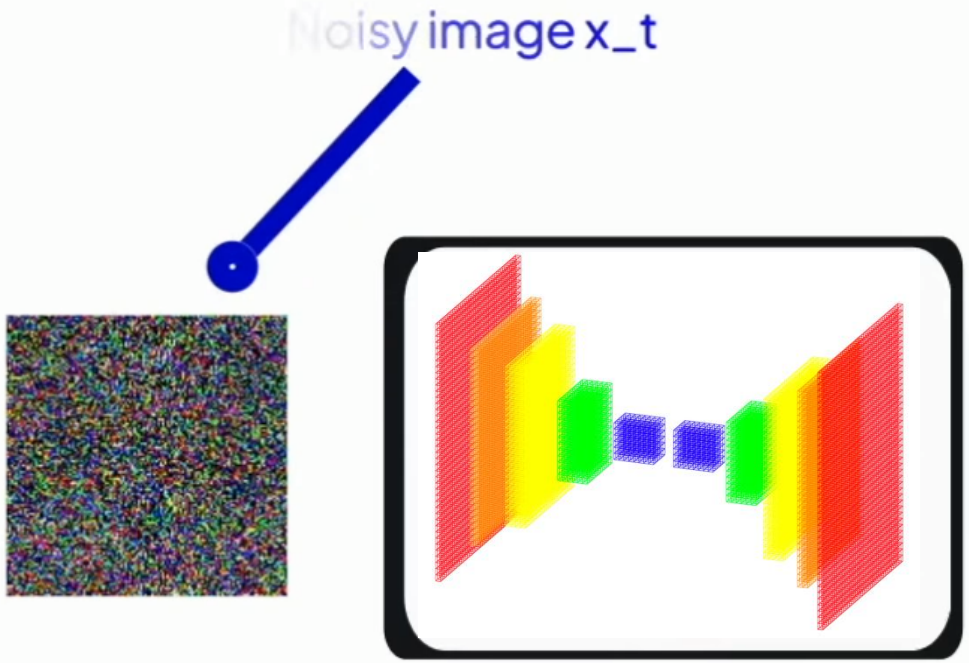
Predict noise and calculate difference to predict image





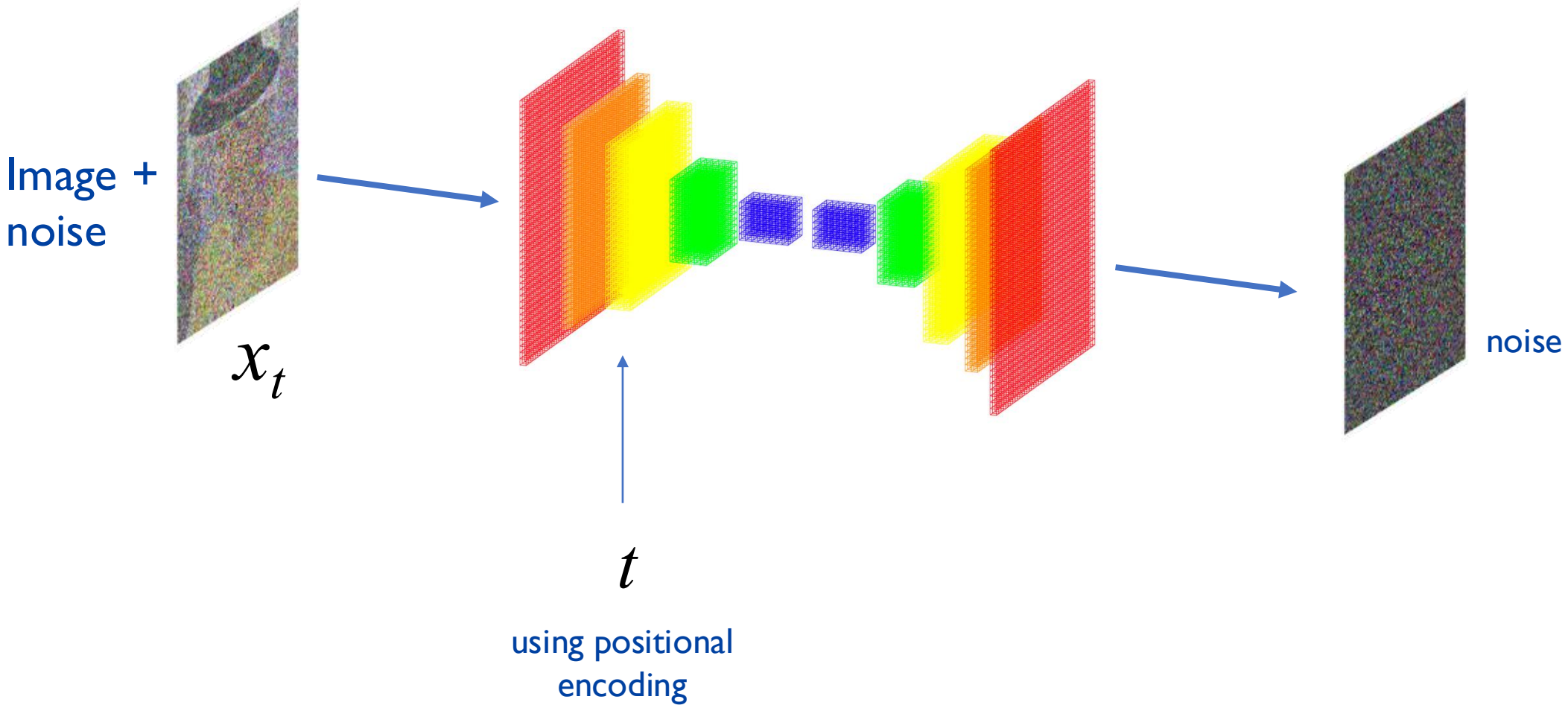


Predicted noise using UNet



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Image Generation – Idea : UNet for noise estimation

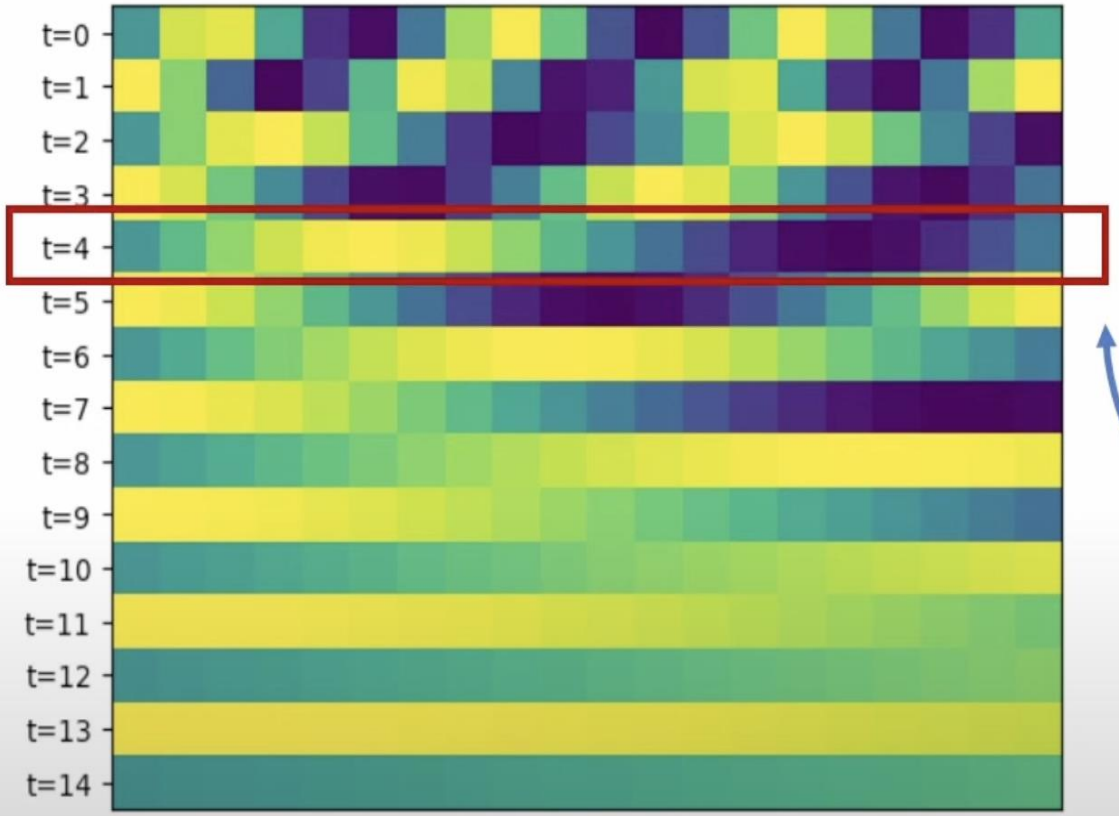


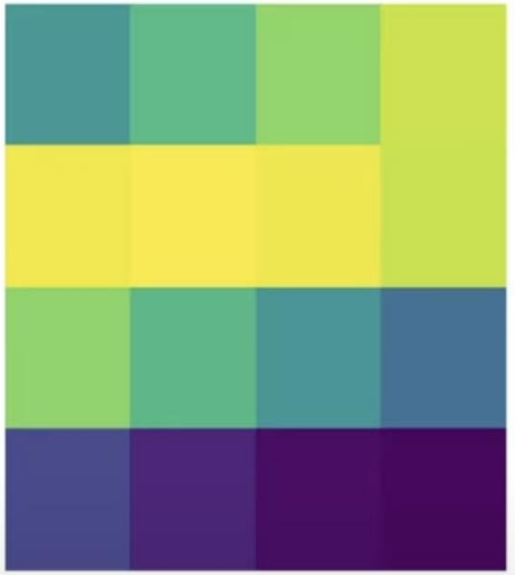
How to include “t” in the Unet?

TIMESTAMP EMBEDDINGS

These are sinusoidal embeddings that are similar to positional embeddings in transformers.
Basically they are used to uniquely encode each timestep t = 1 to N.

Input
t = 4

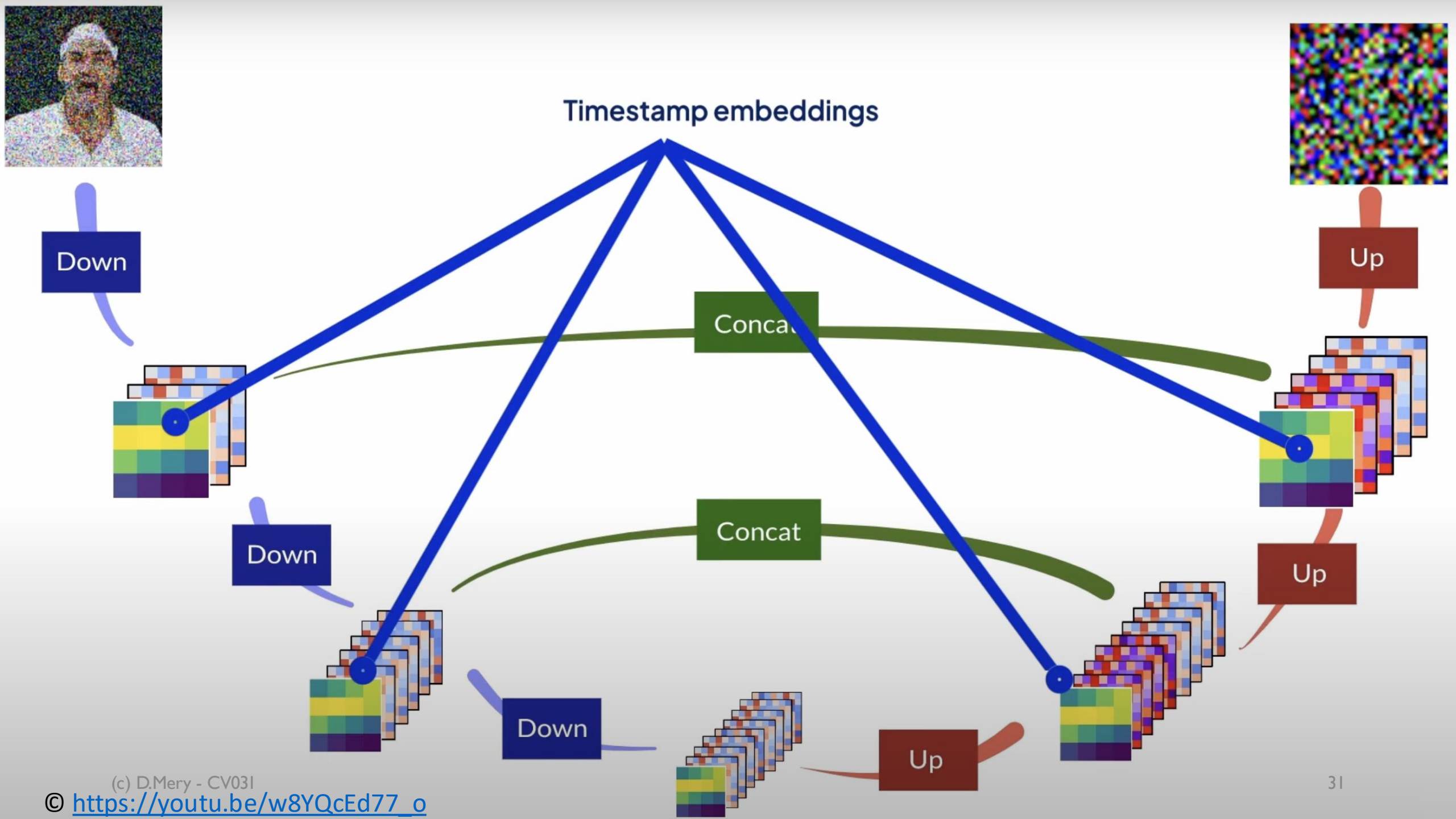




Reshaped Timestamp Embeddings
To make it 2-Dimensional

Lookup the embedding for t = 4

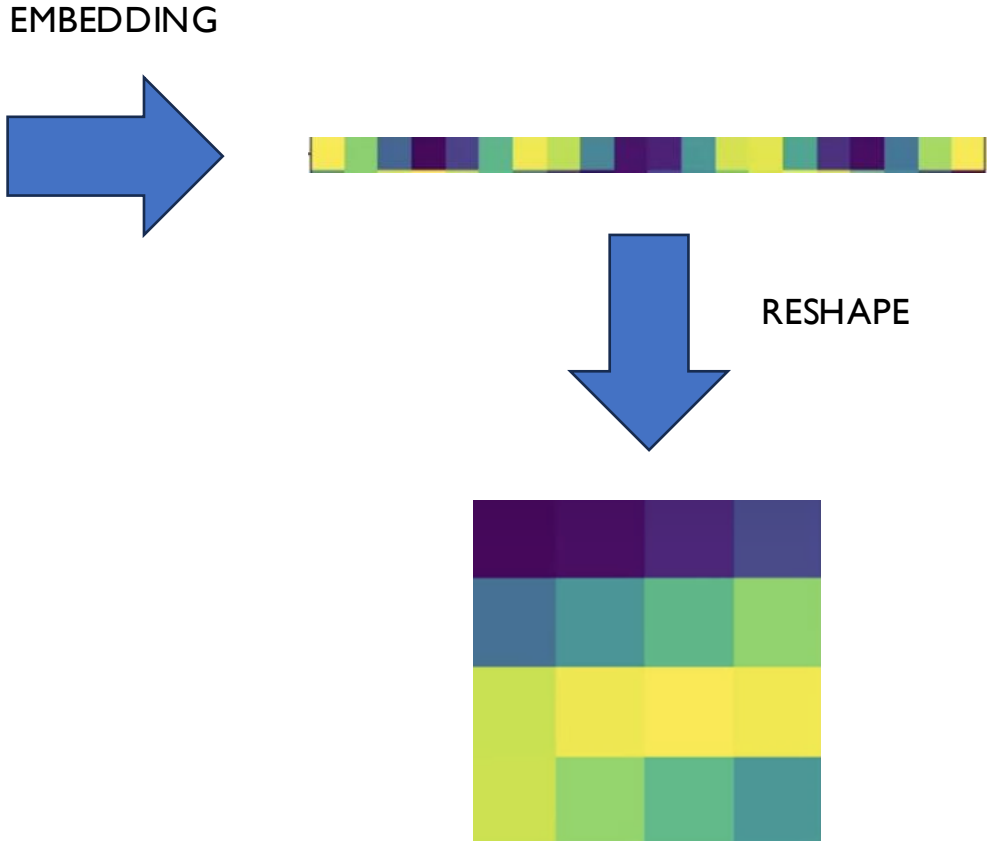
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How to include text in the UNet to generate images?

"Man screaming"





Appending fixed length text embeddings obtained from a network



Down

Down

Down

Concat

Concat

Up

Up

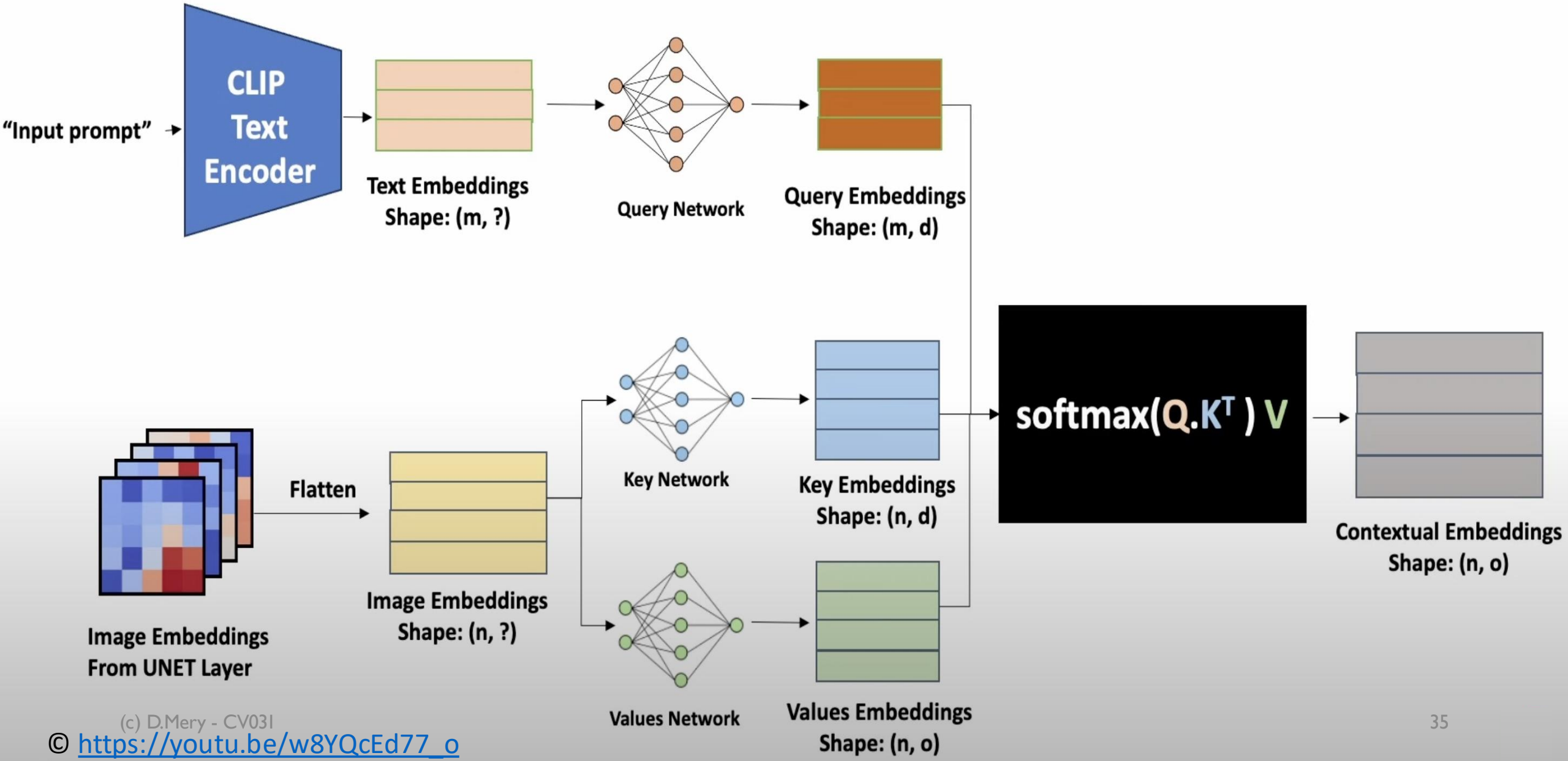
Up

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How to include text in the Unet to generate images?

A better ideal with Attention

TEXT TO IMAGE ATTENTION LAYER



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32x32x3

Down

16x16x8

"Man screaming"

Attn Down

8x8x16

Down

4x4x32

Up

8x8x32

Concat

Concat

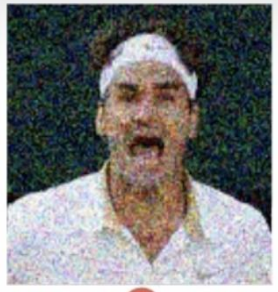
"Man screaming"

Attn Up

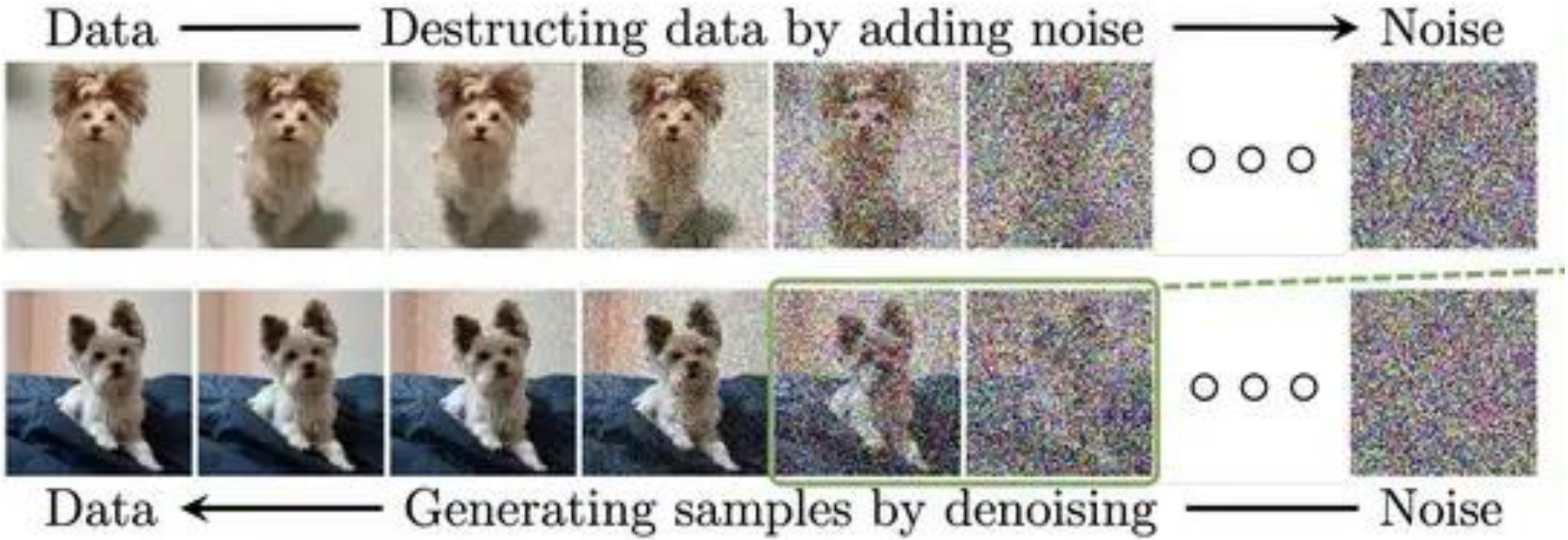
16x16x16

Up

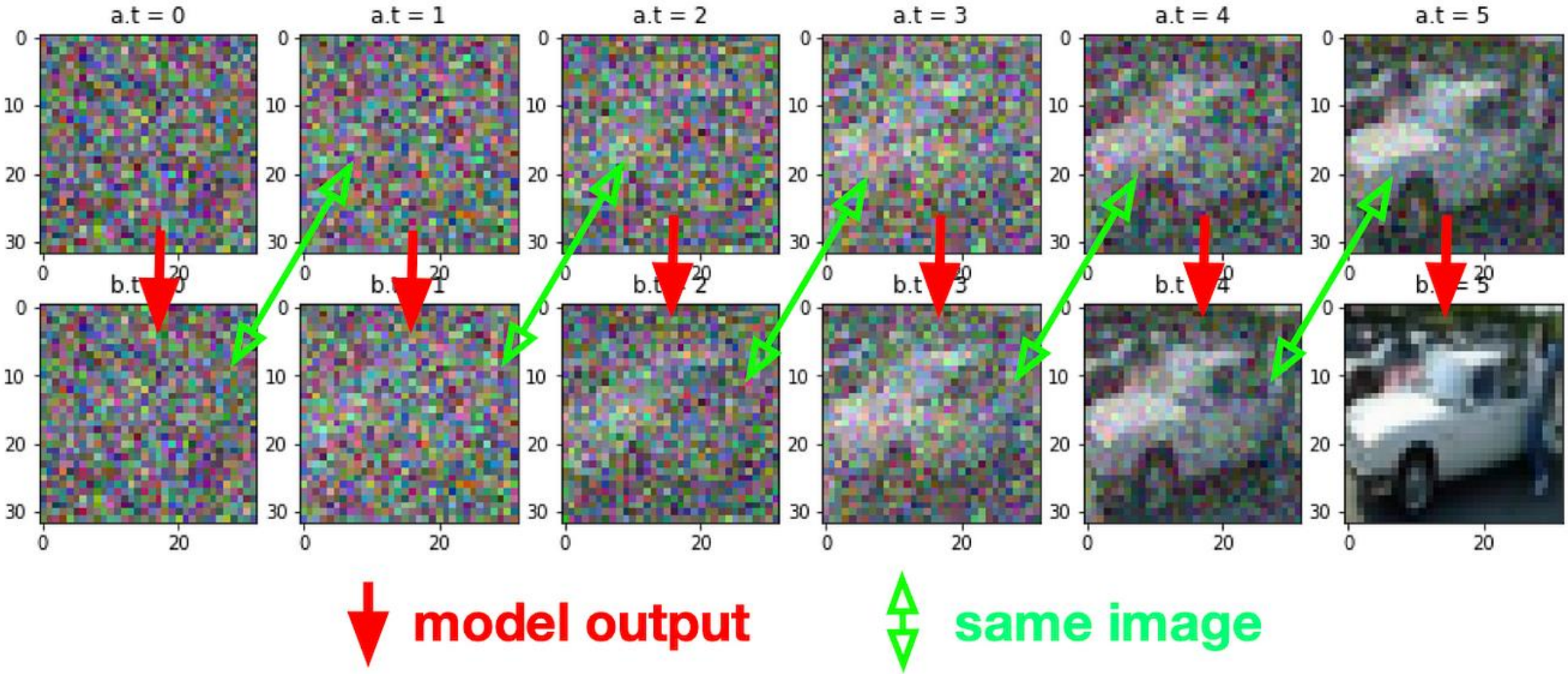
32x32x3



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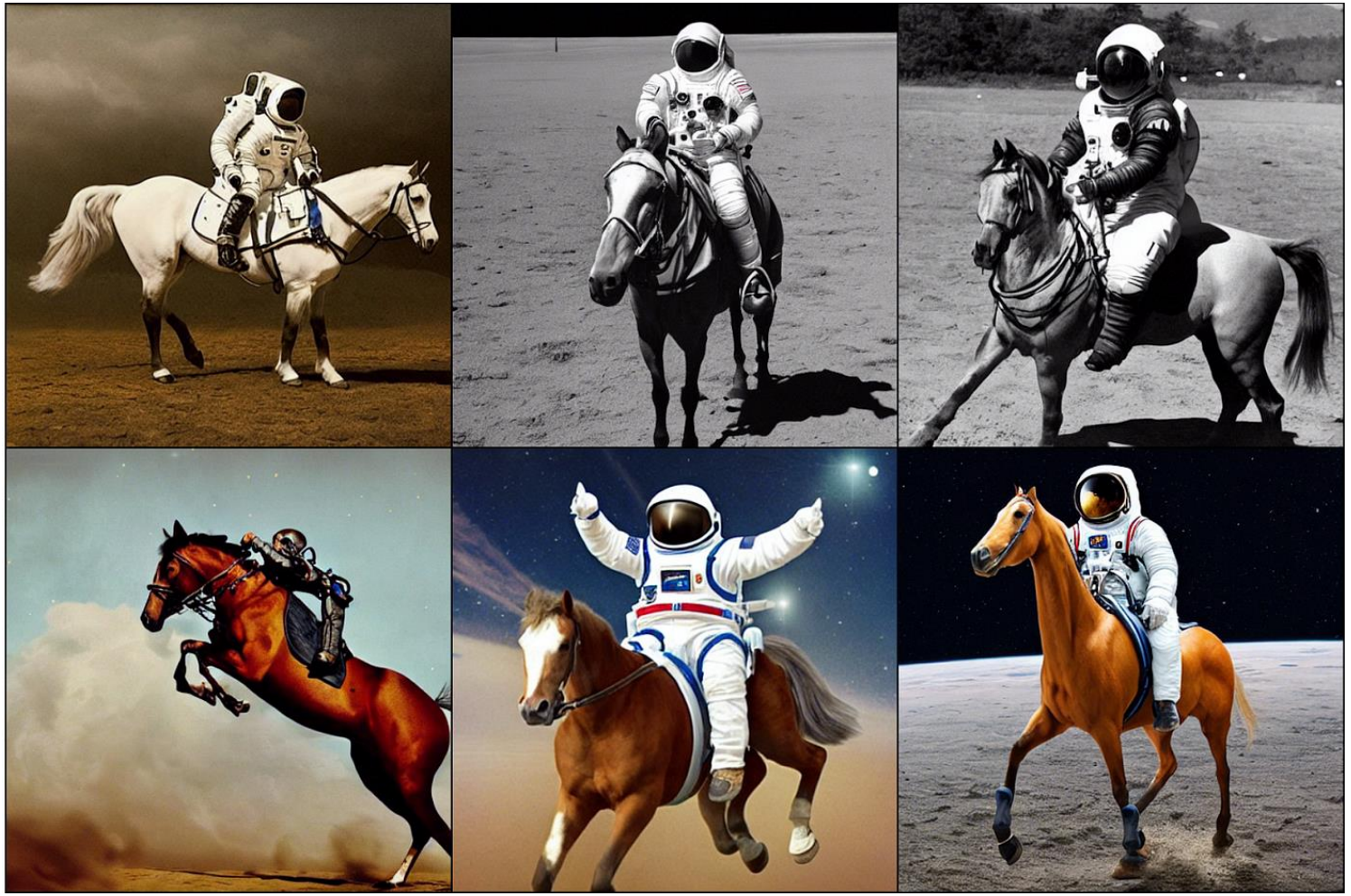


https://www.superannotate.com/blog/diffusion-models



https://tree.rocks/make-diffusion-model-from-scratch-easy-way-to-implement-quick-diffusion-model-e60d18fd0f2e

Prompt: a photograph of an astronaut riding a horse



https://klaviyo.tech/trying-to-understand-stable-diffusion-ea55eefca798

Prompt: a photograph of an astronaut riding a horse

